CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 39-2020: A 29-Month-Old Boy with Seizure and Hypocalcemia

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PRESENTATION OF CASE

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Dr. Neil D. Fernandes: A 29-month-old boy was transferred to this hospital because of a seizure and hypocalcemia.

Approximately 2 months before the current evaluation, the patient had a febrile seizure associated with an acute viral illness. After the seizure, his mother reported that his "waddling" gait had become more pronounced. Two weeks before the current evaluation, the patient was seen in the orthopedic clinic at another hospital. On examination, the cover-up test — in which the distal lower leg is covered and the alignment of the proximal lower leg is compared with that of the upper leg — was negative, a finding consistent with physiologic bowing of the leg. There was a pes planus (flatfoot) deformity that resolved when the patient was not bearing weight on the foot, and there was an external (positive) foot progression angle with ambulation. The mother was counseled that the findings were within the normal range for gait and musculoskeletal development. One day before the current evaluation, the mother noticed that the child's legs "locked up" and appeared "shaky" when he was walking over a threshold.

On the morning of the current evaluation, the patient's mother noticed that the child was grunting while sleeping, with his eyes rolled upward and his arms stiffened and flexed. After less than 1 minute, the grunting and movements stopped and the patient vomited. Emergency medical services were called, and the patient was transported to the emergency department of another hospital. Vital signs, mental status, and findings on physical examination were reportedly normal. Laboratory evaluation was notable for a blood calcium level of 5.0 mg per deciliter (1.25 mmol per liter; reference range, 8.5 to 10.5 mg per deciliter [2.12 to 2.62 mmol per liter]). Intravenous calcium gluconate was administered. The patient was transferred to the emergency department of this hospital, arriving approximately 18 hours after a presumed seizure.

Further history was obtained from the patient's parents. The patient had been delivered at full term and had reportedly met developmental milestones for his age. His medical history was notable for atopic dermatitis, which had been diagnosed when he was 9 weeks of age, as well as multiple environmental and food allergies. One year earlier, evaluation by an allergist had revealed elevated levels on IgE tests specific to food allergens (milk, cashews, pistachios, egg whites, almonds, soybeans, chickpeas, green peas, lentils, peanuts, and sesame seeds) and environmental allergens (dogs, cats, and dust mites), although the patient later underwent food challenges to peanuts, legumes, almonds, and sesame seeds in an allergy clinic and had no symptoms. There were no known drug allergies. The patient had received diphenhydramine, hydroxyzine, epinephrine, and topical betamethasone as needed. He had not received any routine childhood vaccinations because of his parents' refusal.

The patient lived with his mother, father, two older siblings, and dog in New England. Although the mother was a vegetarian, she occasionally included chicken in the patient's "natural" and organic diet. The patient's mother had obsessive–compulsive disorder and anxiety, and his father had oral allergy syndrome and a history of varus knee deformity in childhood. The patient's two siblings had atopic dermatitis, environmental allergies, and food allergies. There was no family history of seizure.

On physical examination, the patient appeared well and was alert and interactive. The temperature was 36.9°C, the height 83.5 cm (3rd percentile for his age), and the weight 12.4 kg (24th percentile for his age). The blood pressure was 95/63 mm Hg, the pulse 114 beats per minute, the respiratory rate 24 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. Muscle spasms occurred in the arms and legs with passive movement. Chvostek's sign (contraction of the facial muscles) was not elicited after tapping on the region of the facial nerve anterior to the external auditory canal; Trousseau's sign (carpal spasm) was not elicited after exertion of pressure from inflation of a sphygmomanometer on the upper arm. Muscle power, bulk, tone, and deep-tendon reflexes were normal, as was the remainder of the examination.

The blood calcium level was 5.6 mg per deciliter (1.40 mmol per liter), the ionized calcium level 0.88 mmol per liter (reference range, 1.14 to 1.30), and the alkaline phosphatase level 673 U per liter (reference range, 142 to 335). Other laboratory test results are shown in Table 1. The patient was admitted to the pediatric intensive care unit.

A diagnostic test was performed.

Table 1. Laboratory Data.*

Variable	Reference Range, Age-Adjusted†	On Admission, This Hospital
Blood		
Sodium (mmol/liter)	135–145	140
Potassium (mmol/liter)	3.4–5.0	4.1
Chloride (mmol/liter)	98–108	104
Carbon dioxide (mmol/liter)	23–32	20
Anion gap (mmol/liter)	3–17	16
Urea nitrogen (mg/dl)	5–20	17
Creatinine (mg/dl)	0.30–1.00	0.25
Glucose (mg/dl)	70–110	106
Calcium (mg/dl)	8.5–10.5	5.6
Ionized calcium (mmol/liter)	1.14–1.30	0.88
Magnesium (mg/dl)	1.7–2.4	1.9
Phosphorus (mg/dl)	4.5–5.5	4.3
Alkaline phosphatase (U/liter)	142–335	673
Parathyroid hormone (pg/ml)	10–60	177
Urine		
Creatinine (mg/dl)	Not defined	46
Calcium (mg/dl)	Not defined	19.7
Phosphate (mg/dl)	Not defined	47.8

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are age-adjusted and for patients who do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

IMAGING STUDIES

Dr. Ruth Lim: Radiographs of the knees (Fig. 1A and 1B) showed several abnormalities that affected the knees symmetrically. The bones were diffusely radiolucent with thinned cortexes; these features suggest generalized demineralization. No acute or healing fracture was detected. The metaphyses of the distal femur and the proximal tibia had a widened, flared contour bilaterally; in addition, the medial aspect of the proximal tibial metaphyses showed a beaked (pointed) contour. The zone of provisional calcification — the area located between the metaphysis and the radiolucent primary growth plate (physis), where chondrocytes are responsible for longitudinal growth of long bones[1] — was frayed in appearance, with

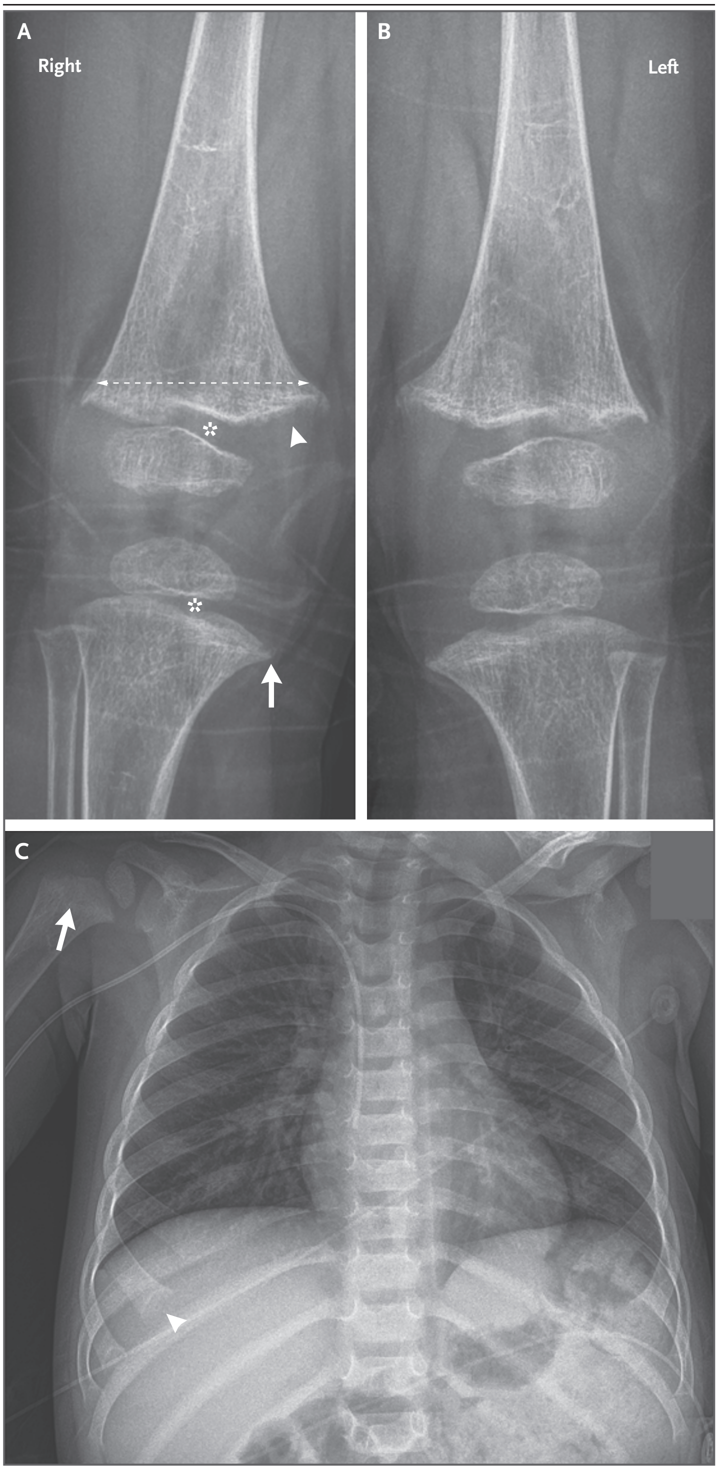


Figure 1. Imaging Studies.

Anteroposterior radiographs of the knees obtained during the current admission (Panels A and B) show generalized demineralization. The metaphyses throughout both knees show flaring (dashed arrow) and fraying and cupping (arrowhead). There is beaking of the medial aspect of the proximal tibial metaphyses (arrow). The radiolucent physes are widened (asterisks). (Abnormalities are marked on the image of the right knee only.) An anteroposterior radiograph of the chest obtained during a previous admission (Panel C) shows subtle flaring and cupping of the right proximal humeral metaphysis (arrow) and mild flaring of the anterior ends of the ribs (arrowhead). The imaging findings are characteristic of rickets, with manifestations seen along the metaphyseal side of the physis at the sites of most rapid bone growth (i.e., the knee, anterior ends of the ribs, wrist, proximal humerus, and distal tibia). Failure of mineralization leads to disorganized chondrocyte growth, and hypophosphatemia leads to impaired apoptosis of hypertrophic chondrocytes. The result is widening of the physis and fraying and cupping of the metaphysis. A peripherally inserted central catheter is also present.

focal areas of cupping (concavity). The radiolucent physes were abnormally widened in the distal femur and the proximal tibia bilaterally.

Because abnormalities were observed on radiographs of the knees, a previously acquired chest radiograph (Fig. 1C) was reviewed for more information. There were subtle findings, including flaring and cupping of the metaphysis of the right proximal humerus and mild flaring of the anterior ends of the ribs.

DIFFERENTIAL DIAGNOSIS

Dr. Yamini V. Virkud: In this 29-month-old boy with a history of eczema and environmental and food allergies, a febrile seizure had occurred 2 months before the current presentation. Since this seizure, the family had noticed ongoing mild limb and gait abnormalities. On the current presentation, the patient had signs of neuromuscular irritability or tetany, including leg locking and shaking, in addition to seizures, findings that are all consistent with clinically significant hypocalcemia. The presence of this condition was confirmed by low levels of calcium and ionized calcium. A primary question arises from this case: How did such severe hypocalcemia develop in this relatively healthy 29-month-old boy?

HYPOCALCEMIA

The differential diagnosis of hypocalcemia can be divided into causes associated with elevated parathyroid hormone (PTH) levels and causes associated with depressed PTH levels.[2,3] PTH secretion regulates the calcium level and is stimulated by a low blood calcium level.[2,4] PTH acts on bones by triggering the release of calcium and phosphate and acts on the kidneys by triggering the resorption of calcium and excretion of phosphate. It also triggers renal production of 1,25-dihydroxyvitamin D, which increases intestinal calcium and phosphate absorption. As the blood calcium level rises, PTH production is suppressed. Thus, the appropriate homeostatic response to hypocalcemia is secondary hyperparathyroidism due to compensatory responses of the calcium–PTH–vitamin D axis.

HYPOCALCEMIA AND A LOW PTH LEVEL

Causes of hypocalcemia in the context of hypoparathyroidism can be divided into congenital and acquired causes.[5] Congenital causes primarily include genetic disorders that result in abnormal PTH synthesis, such as familial isolated hypoparathyroidism, 22q11.2 deletion syndrome (DiGeorge syndrome), or certain mitochondrial disorders. Acquired causes include infection (e.g., human immunodeficiency virus infection), autoimmune disorders (e.g., autoimmune polyglandular syndrome type 1), destruction of the parathyroid gland after surgery or radiation therapy, and infiltrative causes (e.g., cancer, granuloma, amyloidosis, sarcoidosis, or iron or copper deposition). In this patient, the PTH level was elevated, so these causes of hypocalcemia can be ruled out.

HYPOCALCEMIA AND A HIGH PTH LEVEL

Causes of hypocalcemia in the context of hyperparathyroidism can be divided into those associated with PTH resistance, those associated with calcium loss, and those associated with poor calcium intake or absorption.[3] PTH resistance can be due to end-organ resistance to PTH (pseudohypoparathyroidism), missense mutations in PTH that affect its function, or hypomagnesemia. The loss of PTH-mediated excretion of phosphorus from the kidneys results in an elevated phosphorus level; this patient's slightly low phosphorus level and normal magnesium level rule out PTH resistance.

Calcium loss can result from a variety of acute and chronic causes.[3] Acute diseases such as sepsis, pancreatitis, tumor lysis syndrome, and respiratory alkalosis can result in a low ionized calcium level but would not typically affect the total calcium level. Chronic causes of calcium loss include renal disease, hyperphosphatemia, and osteoblastic metastases. Given this patient's low phosphorus level and the absence of a supporting history, calcium loss would not adequately explain his presentation.

Poor calcium intake or absorption could explain this patient's hypocalcemia. A common cause of poor calcium uptake and poor management of calcium homeostasis is deficiency of or resistance to vitamin D.[6] The diagnosis of vitamin D deficiency and associated rickets fits perfectly with the patient's signs and limb abnormalities and with the development of hypocalcemia and seizures. He also had poor growth, an elevated PTH level, a low phosphorus level, and an elevated alkaline phosphatase level, findings that are all consistent with vitamin D deficiency.

VITAMIN D DEFICIENCY

Vitamin D deficiency can be divided into genetic, prenatal or perinatal, and childhood causes.[7] Genetic disorders that alter the metabolism of vitamin D include deficiencies in enzymes (25-hydroxylase or 1α-hydroxylase) and defects in the vitamin D receptor, which cause vitamin D resistance; however, these disorders seem unlikely, given their absence in the family history. Prenatal and perinatal causes include maternal vitamin D deficiency, prematurity, and exclusive breast-feeding without vitamin D supplementation, but this patient's history does not support these possibilities. Childhood causes include obesity, malabsorption, and low sun exposure, as well as decreased nutritional intake, which seems to be the mostly likely explanation in this patient's case. He had extensive IgE-mediated food allergies, including allergies to cow's milk and eggs, resulting in a severely restricted diet. Combined with a mostly vegetarian diet, these restrictions result in a loss of all sources of vitamin D (fish, eggs, and fortified milk). A typical, picky toddler diet may not contain nondairy sources of calcium (beans and green leafy vegetables), thereby exacerbating the poor calcium absorption caused by vitamin D deficiency. I suspect that the most likely diagnosis in this case is vitamin D deficiency due to poor nutrition associated with food allergies and avoidance. To establish this diagnosis, I would obtain a 25-hydroxyvitamin D level.

DR. YAMINI V. VIRKUD'S DIAGNOSIS

Vitamin D deficiency due to poor nutrition associated with food allergies and avoidance.

Table 2. Diagnostic Testing.*

Variable	Reference Range, Age-Adjusted†	On Admission	Hospital Day 7	Hospital Day 12
Calcium (mg/dl)	8.5–10.5	5.6↓	9.0	9.9
Ionized calcium (mmol/liter)	1.14–1.30	0.88↓	1.20	1.27
Phosphorus (mg/dl)	4.5–5.5	4.3↓	4.6	4.9
Alkaline phosphatase (U/liter)	142–335	673↑	566↑	429↑
Parathyroid hormone, intact (pg/ml)	10–60	177↑	135↑	94↑
25-Hydroxyvitamin D (ng/ml)	20–80	4↓	10↓	15↓
1,25-Dihydroxyvitamin D (pg/ml)	24–86	16↓	316↑	—

* To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for 25-hydroxyvitamin D to nanomoles per liter, multiply by 2.496.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are age-adjusted and for patients who do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

PATHOLOGICAL DISCUSSION

Dr. William T. Rothwell: The diagnostic test result in this case was the markedly low 25-hydroxyvitamin D level of 4 ng per milliliter (10 nmol per liter; age-adjusted normal range, 20 to 80 ng per milliliter [50 to 200 nmol per liter]). Vitamin D deficiency can account for each laboratory abnormality summarized in Table 2.

Activated vitamin D increases intestinal calcium absorption, increases calcium resorption from bone, and decreases calcium excretion in urine. It also serves as a negative regulator of PTH. Thus, if sufficient vitamin D substrate had been available in this patient, 25-hydroxyvitamin D and 1,25-dihydroxyvitamin D would have been appropriately synthesized, calcium would have been sourced from bone and the diet, phosphate excretion in urine would have decreased, and PTH and alkaline phosphatase levels would have been normal.

Although 1,25-dihydroxyvitamin D is the active form of the hormone, 25-hydroxyvitamin D should be measured if deficiency is suspected. The blood 25-hydroxyvitamin D level corresponds with vitamin D intake and activity; the half-life of 25-hydroxyvitamin D is 2 to 3 weeks, as compared with a half-life of 4 to 6 hours for 1,25-dihydroxyvitamin D. However, the 1,25-dihydroxyvitamin D level can vary depending on the patient's vitamin D, calcium, phosphorus, and PTH levels. For example, this patient's 1,25-dihydroxyvitamin D level was initially low. Seven days after vitamin D repletion was started, the 1,25-dihydroxyvitamin D level was high. This finding may be misinterpreted to mean that the patient is overly replete with vitamin D, when in fact he remains deficient. Instead, obtaining the 1,25-dihydroxyvitamin D level may be useful for the diagnosis of genetic rickets or in the context of chronic kidney failure, unexplained hypercalcemia, or hypercalciuria.[8]

PATHOLOGICAL DIAGNOSIS

Vitamin D deficiency.

DISCUSSION OF MANAGEMENT

Dr. Deborah M. Mitchell: Nutritional rickets is often thought of as a historical relic — the discovery of vitamin D and the implementation of supplementation and food-fortification strategies in the early 20th century sharply decreased the prevalence of what had been a common disease, particularly in urban centers.[9] However, the incidence of vitamin D deficiency appears to be rising in high-income countries,[10,11] and the prevalence is persistently high in low- and middle-income countries.[12] In a 2015 case-finding survey of Canadian pediatricians, risk factors for nutritional rickets included demographic characteristics (recent immigration), physiological features (skin melanin content, food allergies, feeding challenges associated with prematurity, developmental delay, and maternal vitamin D deficiency), and aspects of diet (lack of vitamin D supplementation and dietary restrictions) and behavior (lack of sun exposure).[13] Of note, this patient had three of these risk factors (food allergies, lack of vitamin D supplementation, and dietary restrictions).

Patients who have profound hypocalcemia that is complicated by severe symptoms (seizure or tetany) receive intravenous calcium to rapidly restore normocalcemia. Unfortunately, there is limited evidence from randomized, controlled trials to guide subsequent repletion of vitamin D stores. In 2016, an international panel developed recommendations for the prevention and management of nutritional rickets; doses and duration of therapy are outlined in Table 3.[14] The panel recommended a daily dosing strategy to minimize the risk of hypercalcemia but also recognized that a single high dose may be more feasible in some circumstances and thus offered dosing recommendations for this strategy, as well. Vitamin D_2 (ergocalciferol) and vitamin D_3 (cholecalciferol) have similar efficacy when given in daily doses; some evidence suggests that vitamin D_3 has better efficacy than vitamin D_2 when given in a single dose.[15] In addition, adequate calcium intake, through dietary calcium or supplementation, is necessary to support bone remineralization. The current recommendations suggest intake of at least 500 mg of calcium daily,[14] although one trial suggests that intake of 1000 mg daily promotes more rapid healing of rickets.[16] Adequate calcium provision is also necessary to prevent hungry bone syndrome, a condition in which there is paradoxical worsening of hypocalcemia after initiation of vitamin D therapy. This condition is thought to be due to rapid bone-mineral uptake by the demineralized skeleton, and management may require very high doses of calcium, as well as occasional therapeutic doses of calcitriol (1,25-dihydroxyvitamin D), the active form of vitamin D.[17]

The response to treatment can be monitored by obtaining blood levels of calcium, phosphate, 25-hydroxyvitamin D, PTH, and alkaline phosphatase and the urinary calcium:creatinine ratio approximately 4 weeks after the initiation of treatment and at monthly intervals until resolution of laboratory abnormalities.[18] In most children with nutritional rickets, laboratory values will typically normalize and radiographs will show substantial healing after 3 months of adequate treatment, although some patients may require continued treatment.[14] Ongoing secondary prevention with daily maintenance doses of vitamin D (as shown in Table 3) and continued provision of at least 500 mg of calcium daily should then be initiated and continued indefinitely.

Table 3. **Treatment Doses of Vitamin D.***

Age	Daily Treatment Dose, 90 Days	Single Treatment Dose	Daily Maintenance Dose
		international units	
<3 mo	2000	NA	400
3 to 12 mo	2000	50,000	400
>12 mo to 12 yr	3000–6000	150,000	600
>12 yr	6000	300,000	600

* Adapted from Munns et al.[14] NA denotes not applicable.

Discussion of Food Allergy

Risks of Food Avoidance

Dr. Virkud: Food avoidance is often considered to be a benign intervention, but this case shows that even medically necessary food restrictions can have severe consequences. For patients with a definite IgE-mediated food allergy, food avoidance is medically necessary and yet still has implications that require monitoring. Many studies have documented decreased growth in both weight and height in patients with IgE-mediated and non–IgE-mediated food allergies, and the largest study showed that children with a milk allergy were particularly susceptible to decreased weight.[19,20]

The restricted diets of children with food allergies not only affect growth but also can lead to specific nutritional deficiencies. In this patient, the combination of food restrictions and a vegetarian diet led to a loss of all common sources of vitamin D and calcium. Other micronutrient deficiencies reported in children with food restrictions include deficiencies in iodine, fatty acids such as n−3 and n−6 (also known as omega-3 and omega-6, respectively), vitamin A, zinc, and B vitamins (Fig. 2).[20-22] Children with food allergies can also be picky eaters, so once common allergens such as milk, eggs, nuts, or seafood are removed from the diet, the risk for nutritional deficiencies increases dramatically. Parents may need extra help from dietitians, who can develop strategies for incorporating nutritionally replete foods into the diet.

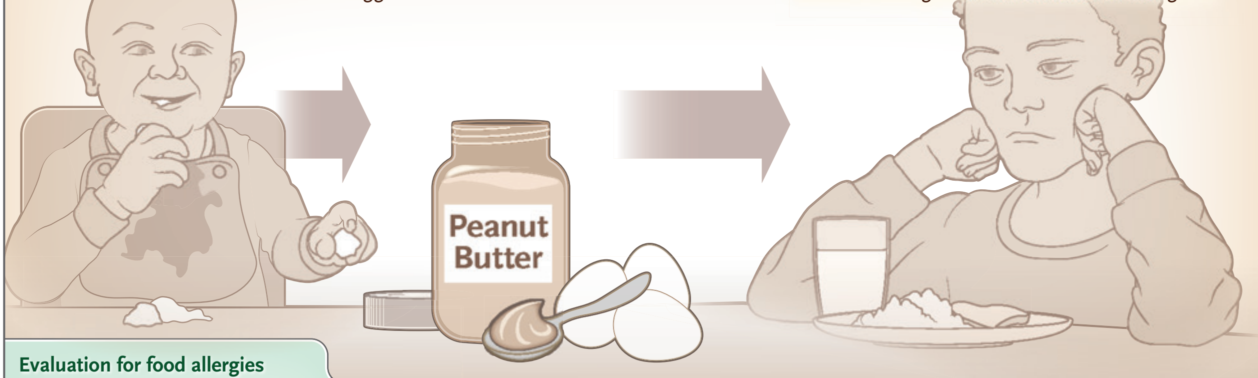
Avoiding Misdiagnosis of Food Allergies

The prevalence of IgE-mediated food allergies and of hospitalizations for food-triggered anaphylaxis have been increasing over the past few decades, but accurate estimates of prevalence are biased by the misdiagnosis of food allergies.[23]

Introduction of food allergens

Early introduction of common allergens is recommended; data are strongest for peanuts and eggs.

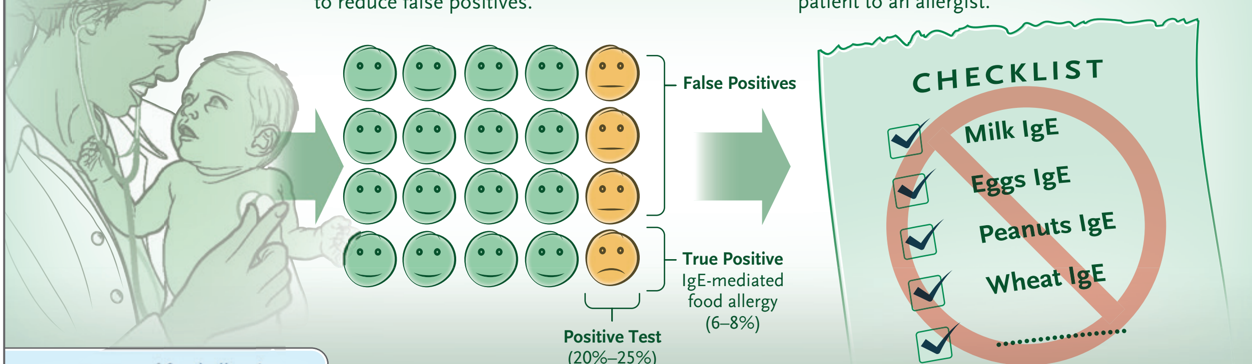
First-time parents, anxious parents, and parents of picky eaters may need more guidance and creative strategies for introduction of allergens.



Evaluation for food allergies

Screening tests for allergies have high false positive rates in the general pediatric population. Obtaining a detailed clinical history is essential to reduce false positives.

Avoid allergen panels; conduct testing only if the patient has a history of an allergic reaction with a clear food trigger. When in doubt, refer the patient to an allergist.



Management of food allergies

Evaluate the patient's growth and diet. If there are concerns, refer the patient to a dietitian.

Nutrient	Milk	Eggs	Nuts and seeds	Fish	Other sources
Vitamin D	✓	✓		✓	
Calcium	✓				Beans and green leafy vegetables
Iodine	✓				Meat, poultry, fruits, vegetables, and table salt
N–3 and N–6 Fatty Acids	✓	✓	✓	✓	Poultry and whole grains
Vitamin A	✓	✓			Liver, kidney, sweet potatoes, and carrots
Zinc	✓	✓	✓		Oysters, liver, beef, chicken, and rice
B Vitamins	✓	✓		✓	Meat

Encourage regular visits to an allergist to assess for outgrown allergies.

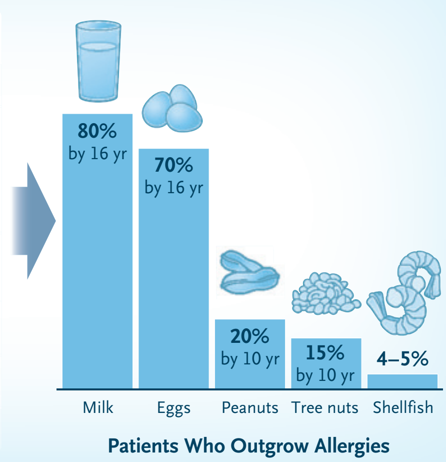


Figure 2. Lessons about IgE-Mediated Food Allergy.

The most accurate diagnostic tool is an oral food challenge, in which a patient with a food allergy is fed gradually increasing doses of the suspected allergen.[24] However, this procedure is time- and resource-consuming and poses a risk for allergic reactions, and therefore the diagnosis of food allergies often relies instead on clinical history, skin-prick testing, and allergen-specific IgE blood testing. Clinical and laboratory testing is severely limited by poor specificity. On average, skin-prick testing has higher sensitivity and specificity than allergen-specific IgE testing; for both tests, the overall sensitivity ranges from 55 to 96% and the specificity ranges from 38 to 73%, depending on the allergen tested.[24] However, on the basis of a study of samples obtained from the general pediatric population from the Third National Health and Nutrition Examination Survey, approximately 20 to 25% of children have positive IgE blood tests specific to food allergens, even though the true prevalence of IgE-mediated food allergy is likely to be closer to 6 to 8%.[25-27] This translates to a high false positive rate associated with the use of allergen-specific IgE testing for screening of the general population (Fig. 2).

In this case, the presence of positive tests specific to milk, eggs, tree nuts, peanuts, and legumes, followed by the successful reintroduction of some of these foods, is a classic example of the limitations of diagnostic testing in the allergy field. A key method for reducing misdiagnosis of food allergies involves avoiding indiscriminate use of IgE blood testing in screening for food allergies. Panels of allergen-specific IgE are particularly problematic, because they often uncover false positives and lead to unnecessary food avoidance.[24] After a thorough clinical history of the suspected allergic reaction has been obtained, IgE and skin-prick testing can be used as confirmatory tests, targeting only food allergens that are suspected to trigger the reaction. Given the time that it takes to get an appointment with an allergist as a new patient, to undergo all necessary screening tests, and then to reach the top of a long wait list for a confirmatory oral food challenge, it can take over a year for a false positive IgE test to be resolved. During this time, the patient is at risk for the development of nutritional deficiencies, as well as anxieties regarding food allergies, and may actually even have an increased risk of developing a new food allergy.

Another method for reducing the prevalence of food allergies includes close follow-up with an allergist to identify outgrown allergies. Rates of outgrown allergies are the highest for milk and eggs (approximately 70% and 80%, respectively) and are lower for nuts and seafood (Fig. 2).[28] To monitor for the possibility of an outgrown allergy, allergists conduct repeat skin-prick and IgE testing. On the basis of the results, allergists guide reintroduction of the food, either by oral food challenge in the clinic or introduction at home.[28] Many patients who outgrow an allergy have intense anxiety about reintroducing the food, and allergists and psychologists who specialize in food allergies can help guide this process.[29] For patients with food allergies, yearly follow-up with an allergist is essential for reintroducing foods as early as possible in order to reduce potential risks associated with an unnecessarily restricted diet.

Pediatricians and allergists can play an important role in decreasing risks by encouraging early introduction of allergens to reduce the likelihood of developing an allergy, by avoiding screening IgE tests that often uncover false positive results, by closely monitoring patients with restricted diets to ensure they are supplementing to avoid micronutrient deficiencies, and by following patients regularly to clear any food-allergy diagnoses that have been outgrown. Dietitians and psychologists who specialize in food allergies can be invaluable in supporting these efforts. Finally, all clinicians can play a role in reducing parental misconceptions and fears about food allergies that often lead to delayed food introduction in infants and by encouraging families to reach out to their pediatricians and allergists for guidance.

FOLLOW-UP

Dr. Fernandes: Intravenous calcium gluconate was administered until the ionized calcium level was more than 1 mmol per liter, and the patient also received oral calcium supplements. We had difficulty maintaining the ionized calcium level at more than 1 mmol per liter for sustained durations, so a continuous calcium infusion was administered for 7 days. For his underlying diagnosis of vitamin D deficiency, he was given vitamin D at a dose of 50,000 IU on day 1, followed by 1000 IU per day, as well as elemental calcium at a dose of 300 mg three times per day. The patient was ultimately discharged while receiving calcium carbonate and vitamin D. At the time of discharge, the calcium level was 9.4 mg per deciliter (2.35 mmol per liter), the ionized calcium level 1.25 mmol per liter, the alkaline phosphatase level 429 U per liter, and the PTH level 94 pg per milliliter. At his most recent visit to the endocrine clinic, he was doing well and his laboratory values had normalized. At 3 years 11 months of age, his height was in the 19th percentile and his weight was in the 69th percentile. He had passing results on food challenges for several allergens that were performed in the office, but his intake of milk, eggs, and multiple tree nuts continues to be restricted.

FINAL DIAGNOSIS

Vitamin D deficiency.

This case was presented at the "Pediatrics Primary Care" postgraduate course.

Dr. Mitchell reports receiving consulting fees from Chugai and Amolyt; and Dr. Rothwell, being cofounder and owning shares in Surmount and holding pending patent US20180339065A1 on intrathecal administration of adeno-associated viral vectors for gene therapy in neurodegenerative diseases and pending patent US20170043035A1 on methods and compositions for treating metastatic breast cancer and other cancers in the brain with adeno-associated viral vectors for gene therapy, previously licensed to Biogen, for which royalties were previously received. No other potential conflict of interest relevant to this article was reported.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

1. Chang CY, Rosenthal DI, Mitchell DM, Handa A, Kattapuram SV, Ambrose JH. Imaging findings of metabolic bone disease. Radiographics 2016;36:1871-87.
2. Cooper MS, Gittoes NJL. Diagnosis and management of hypocalcaemia. BMJ 2008;336:1298-302.
3. Root A. Disorders of mineral homeostasis in children and adolescents. Oncohema Key 2014 (https://oncohemakey.com/disorders-of-mineral-homeostasis-in-children-and-adolescents/).
4. Underland L, Markowitz M, Gensure R. Calcium and phosphate hormones: vitamin D, parathyroid hormone, and fibroblast growth factor 23. Pediatr Rev 2020;41:3-11.
5. Nelson textbook of pediatrics. 21st ed. New York: Elsevier, 2020 (https://www.us.elsevierhealth.com/nelson-textbook-of-pediatrics-2-volume-set-9780323529501.html).
6. Rajah J, Thandrayen K, Pettifor JM. Clinical practice: diagnostic approach to the rachitic child. Eur J Pediatr 2011;170:1089-96.
7. Misra M, Pacaud D, Petryk A, Collett-Solberg PF, Kappy M. Vitamin D deficiency in children and its management: review of current knowledge and recommendations. Pediatrics 2008;122:398-417.
8. Case Records of the Massachusetts General Hospital (Case 3-2009). N Engl J Med 2009;360:398-407.
9. Wheeler BJ, Snoddy AME, Munns C, Simm P, Siafarikas A, Jefferies C. A brief history of nutritional rickets. Front Endocrinol (Lausanne) 2019;10:795.
10. Thacher TD, Fischer PR, Tebben PJ, et al. Increasing incidence of nutritional rickets: a population-based study in Olmsted County, Minnesota. Mayo Clin Proc 2013;88:176-83.
11. Högberg U, Winbo J, Fellman V. Population-based register study of children born in Sweden from 1997 to 2014 showed an increase in rickets during infancy. Acta Paediatr 2019;108:2034-40.
12. Thacher TD, Fischer PR, Strand MA, Pettifor JM. Nutritional rickets around the world: causes and future directions. Ann Trop Paediatr 2006;26:1-16.
13. Canadian Paediatric Surveillance Program (CPSP). Vitamin D deficiency rickets. 2015 (https://www.cpsp.cps.ca/uploads/surveys/vitamin-d-deficiency-rickets-survey-results.pdf).
14. Munns CF, Shaw N, Kiely M, et al. Global consensus recommendations on prevention and management of nutritional rickets. J Clin Endocrinol Metab 2016;101:394-415.
15. Tripkovic L, Lambert H, Hart K, et al. Comparison of vitamin D2 and vitamin D3 supplementation in raising serum 25-hydroxyvitamin D status: a systematic review and meta-analysis. Am J Clin Nutr 2012;95:1357-64.
16. Thacher TD, Smith L, Fischer PR, Isichei CO, Cha SS, Pettifor JM. Optimal dose of calcium for treatment of nutritional rickets: a randomized controlled trial. J Bone Miner Res 2016;31:2024-31.
17. Thirunagari R, Taha D. SAT-275 "nutritional rickets: still a problem in the 21st century." J Endocr Soc 2019;3:Suppl 1 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6551936/).
18. Lambert AS, Linglart A. Hypocalcaemic and hypophosphatemic rickets. Best Pract Res Clin Endocrinol Metab 2018;32:455-76.
19. Mehta H, Ramesh M, Feuille E, Groetch M, Wang J. Growth comparison in children with and without food allergies in 2 different demographic populations. J Pediatr 2014;165:842-8.
20. Meyer R. Nutritional disorders resulting from food allergy in children. Pediatr Allergy Immunol 2018;29:689-704.
21. Shaker M, Venter C. The ins and outs of managing avoidance diets for food allergies. Curr Opin Pediatr 2016;28:567-72.
22. Institute of Medicine. Dietary reference intakes: the essential guide to nutrient requirements. 2006 (https://www.nap.edu/catalog/11537/dietary-reference-intakes-the-essential-guide-to-nutrient-requirements).
23. Tang MLK, Mullins RJ. Food allergy: is prevalence increasing? Intern Med J 2017;47:256-61.
24. Sicherer SH, Sampson HA. Food allergy: a review and update on epidemiology, pathogenesis, diagnosis, prevention, and management. J Allergy Clin Immunol 2018;141:41-58.
25. McGowan EC, Peng RD, Salo PM, Zeldin DC, Keet CA. Changes in food-specific IgE over time in the National Health and Nutrition Examination Survey (NHANES). J Allergy Clin Immunol Pract 2016;4:713-20.
26. McGowan EC, Keet CA. Prevalence of self-reported food allergy in the National Health and Nutrition Examination Survey (NHANES) 2007-2010. J Allergy Clin Immunol 2013;132:1216.e5-1219.e5.
27. Gupta RS, Springston EE, Warrier MR, et al. The prevalence, severity, and distribution of childhood food allergy in the United States. Pediatrics 2011;128(1):e9-e17.
28. Longo G, Burks AW, Krauss B, Barbi E. IgE-mediated food allergy in children. Lancet 2013;382:1656-64. .
29. Polloni L, Ferruzza E, Ronconi L, et al. Assessment of children's nutritional attitudes before oral food challenges to identify patients at risk of food reintroduction failure: a prospective study. Allergy 2017;72:731-6.